ASSIGNMENT: RECORDED VIDEO PRESENTATION

Chapter II – Parallel Computing Concepts

Course: Elective 9 – Parallel and Distributed Computing
Chapter Covered: Chapter II – Parallel Computing Concepts
Type: Individual Assignment
Mode: Recorded Video Presentation
Submission: Via LMS / STEP S / google drive

I. Assignment Objective

This assignment aims to assess the student's ability to understand, explain, and communicate the fundamental concepts of parallel computing, including its definition, types, models, and performance considerations, using clear and logical explanations suitable for both technical and non-technical audiences.

II. Task Description

Each student shall create a recorded video presentation discussing Parallel Computing Concepts.

The video must demonstrate:

- Conceptual understanding
- Proper use of terminology
- Clear explanation using examples
- Logical flow of ideas

This is an individual task. No group submissions.

III. Required Video Content (Must Follow This Order)

Part 1: Introduction

- State your name, course, and section
- Briefly introduce parallel computing
- Explain why parallel computing is important in modern systems

Part 2: Definition of Parallel Computing

- Define parallel computing in your own words
- Explain how it differs from sequential computing
- Give one simple real-world analogy

Part 3: Types of Parallel Computing

Discuss at least TWO (2) of the following:

- Data Parallelism
- Task Parallelism
- Bit-level Parallelism
- Instruction-level Parallelism

Explain:

- What it is
- Where it is commonly used

Part 4: Parallel Computing Models / Concepts

Discuss at least ONE (1) of the following:

- Shared Memory Model
- Threads and Multithreading
- Parallel Random Access Machine (PRAM)
- Parallel performance concepts (speedup, efficiency – conceptual only)

Part 5: Local / Practical Example

Give one Philippine or real-life example where parallel computing is used, such as:

- Image or video processing
- Gaming systems
- AI applications
- Computer laboratories
- Engineering or scientific simulations

Explain why parallel computing is needed in your example.

Part 6: Conclusion

- Summarize key points learned
- Answer this question:

"Why is parallel computing essential in today's computing environment?"

IV. Video Technical Requirements

- **Length:** 5–10 minutes (strictly enforced)

- **Format:** MP4
- **Orientation:** Landscape
- **Audio:** Clear and audible
- **Camera:** Face must be visible (voice-over only is NOT allowed)
- **Slides:** Optional but encouraged (if used, keep them minimal)

Reading directly from slides or scripts will result in point deduction.

V. Academic Integrity Policy

- The video must be original and recorded by the student.
- Copying scripts, AI-generated narration, or other students' work is strictly prohibited.
- Any form of plagiarism will result in a failing mark for the assignment.

VI. Rubric (100 Points)

Criteria	*Points*
Content accuracy & completeness	40
Clarity of explanation	20
Organization & flow	15
Use of examples (local/real-world)	10
Presentation skills & confidence	10
Video quality & time compliance	5
TOTAL	**100**

VII. Submission Instructions

- File name format:
  Lastname_Firstname_ParallelComputing_Video
- Upload on the designated platform on or before the deadline.
- Late submissions will follow the course penalty policy.

VIII. Reminder to Students

This assignment tests not just what you know, but how well you understand and explain computing concepts, a critical skill for future IT professionals.